Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Mello Ranch

Physical address of dairy:

Number and Street	City	County	Zip Code
27181 E Carter RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X187-X260-XX03-XXXX X187-X260-XX04-XXXX

B. OPERATORS

Mello, Jarred

Operator name: Mello, Jarred

Telephone no.: Landline: ; Cellular: (209) 573-3748

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 22	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Mello, Jarred

Legal owner name: Mello, Jarred

Telephone no.: Landline: ; Cellular: (209) 573-3748

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 22	Escalon	CA	95320

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	850	4,050
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	850	4,050
Average number	0	0	0	0	850	4,050
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 16,990.75 *tons per reporting period*

Total nitrogen from manure: 250,390.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 175,273.00 *lbs per reporting period*

Total phosphorus from manure: 17,706.15 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,578,000 *gallons*

Total nitrogen generated: 19,537.01 *lbs*

Total phosphorus generated: 2,399.38 *lbs*

Total potassium generated: 26,043.86 *lbs*

Total salt generated: 229,893.40 *lbs*

	4,578,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,578,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW #1	Ground water
IW #2	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/26/2023	Solid commercial fertilizer *CAN17*	2.93 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	995.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	995.20	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	Corral solids	564.00 *ton*	Dry-weight	21.7		25,200.00	8,030.00	34,000.00		40.20

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	22,257.24	7,092.29	30,029.62	355,056.05
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	22,257.24	7,092.29	30,029.62	355,056.05

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	17	17	1	none	X187-X260-XX04-XXXX
Field 2	42	42	2	process wastewater	X187-X260-XX03-XXXX X187-X260-XX04-XXXX
Field 3	22	22	1	none	X187-X260-XX03-XXXX
Totals for areas that were used for application	42	42	2		
Totals for areas that were not used for application	39	39	2		
Land application area totals	81	81	4		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/20/2020: Pistachios

Crop: Pistachios Acres planted: 17 Plant date: 10/20/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	0.10	0.10	0.01	0.01	0.00
Total actual harvest content	0.01	0.00	0.00	0.00	0.00

Field 2

Field name: Field 2

Field 2

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 42 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	661.30 *ton*	As-is		62.0	4,790.00	1,260.00	7,930.00		9.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.75	150.84	39.68	249.72	1,165.53

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,502.30 *ton*	As-is		78.0	2,840.00	530.00	3,220.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.77	203.17	37.92	230.35	1,148.90

Field 3

Field name: Field 3

10/20/2020: Pistachios

Crop: Pistachios Acres planted: 22 Plant date: 10/20/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	1.00 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	0.10	0.10	0.01	0.01	0.00
Total actual harvest content	0.05	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/20/2020: Pistachios

Field name: Field 1

Crop: Pistachios

Plant date: 10/20/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,266,500.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	8.50	0.00	0.00	0.00	
Application event totals		8.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,266,500.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,266,500.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	8.50	0.00	0.00	0.00	
Application event totals		8.50	0.00	0.00	0.00	

Field 1 - 10/20/2020: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,266,500.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,266,500.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	8.50	0.00	0.00	0.00	
Application event totals		8.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Field 1 - 10/20/2020: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,292,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Field 2 - 10/31/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.88	4.86	27.14	496.69	504,000.00 *gal*
Application event totals		58.88	4.86	27.14	496.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.88	4.86	27.14	496.69	504,000.00 *gal*
Application event totals		58.88	4.86	27.14	496.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.37	14.05	114.43	711.49	882,000.00 *gal*
Application event totals		89.37	14.05	114.43	711.49	

Field 2 - 06/20/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.37	14.05	114.43	711.49	882,000.00 *gal*
IW #2	Ground water	0.00	0.00	0.00	106.61	4,471,200.00 *gal*
Application event totals		89.37	14.05	114.43	818.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	127.34	5,340,600.00 *gal*
Application event totals		0.00	0.00	0.00	127.34	

Field 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	127.34	5,340,600.00 *gal*
Application event totals		0.00	0.00	0.00	127.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	125.85	5,278,500.00 *gal*
Application event totals		0.00	0.00	0.00	125.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.21	10.33	180.23	1,635.29	966,000.00 *gal*
IW #2	Ground water	0.00	0.00	0.00	103.64	4,347,000.00 *gal*
Application event totals		90.21	10.33	180.23	1,738.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.44	8.98	156.72	1,421.99	840,000.00 *gal*
IW #2	Ground water	0.00	0.00	0.00	100.68	4,222,800.00 *gal*
Application event totals		78.44	8.98	156.72	1,522.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	118.45	4,968,000.00 *gal*
Application event totals		0.00	0.00	0.00	118.45	

Field 3 - 10/20/2020: Pistachios

Field name: Field 3

Crop: Pistachios

Plant date: 10/20/2020

Field 3 - 10/20/2020: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,639,000.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

04/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	8.50	0.00	0.00	0.00	
Application event totals		8.50	0.00	0.00	0.00	

04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,639,000.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,639,000.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

05/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	8.50	0.00	0.00	0.00	
Application event totals		8.50	0.00	0.00	0.00	

05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,639,000.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Field 3 - 10/20/2020: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.24	0.00	0.00	80.82	1,639,000.00 *gal*
Application event totals		4.24	0.00	0.00	80.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	8.50	0.00	0.00	0.00	
Application event totals		8.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Field 3 - 10/20/2020: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

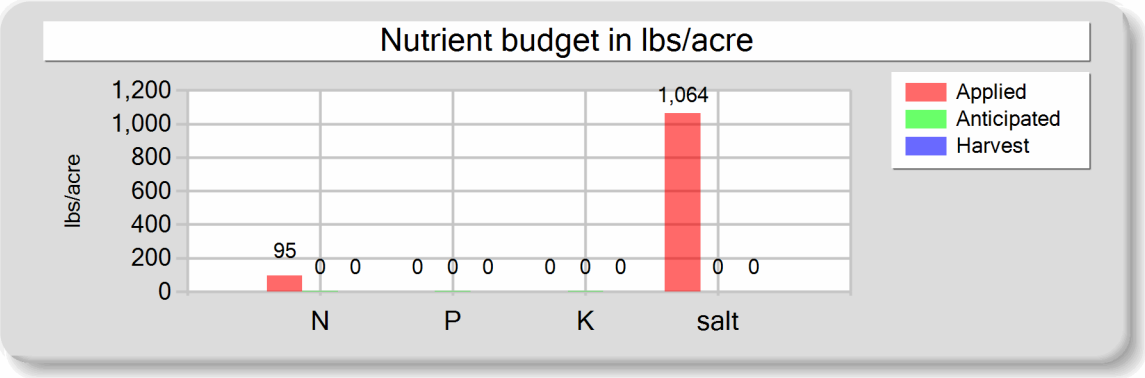
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	4.33	0.00	0.00	82.45	1,672,000.00 *gal*
Application event totals		4.33	0.00	0.00	82.45	

B. NUTRIENT BUDGET

Field 1 - 10/20/2020: Pistachios

Field name: Field 1 Crop: Pistachios Plant date: 10/20/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25.50	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	55.80	0.00	0.00	1,063.70
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	95.30	0.00	0.00	1,063.70
Anticipated crop nutrient removal	0.10	0.01	0.01	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	95.30	0.00	0.00	1,063.70
Applied to removed ratio	0.00	0.00	0.00	0.00

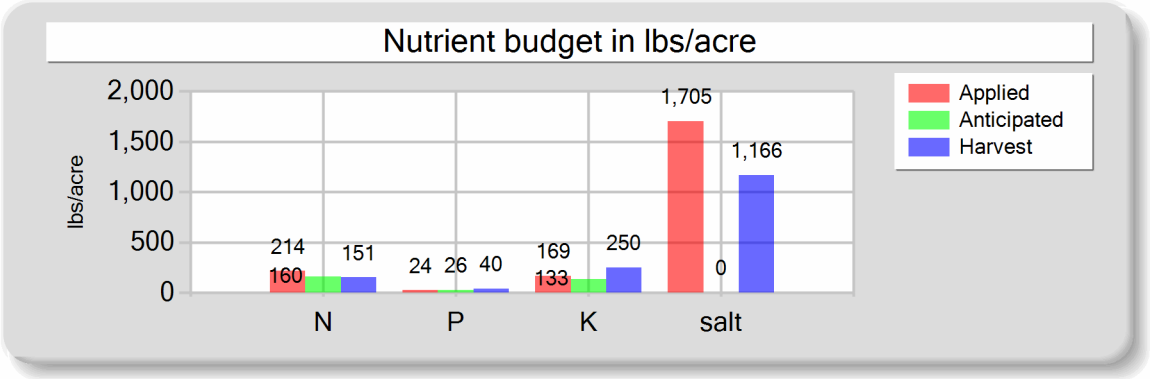
Fresh water applied	
16,668,500.00	*gallons*
613.84	*acre-inches*
36.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/31/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.14	23.77	168.71	1,704.88
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.14	23.77	168.71	1,704.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	150.84	39.68	249.72	1,165.53
Nutrient balance	63.30	-15.91	-81.01	539.36
Applied to removed ratio	1.42	0.60	0.68	1.46

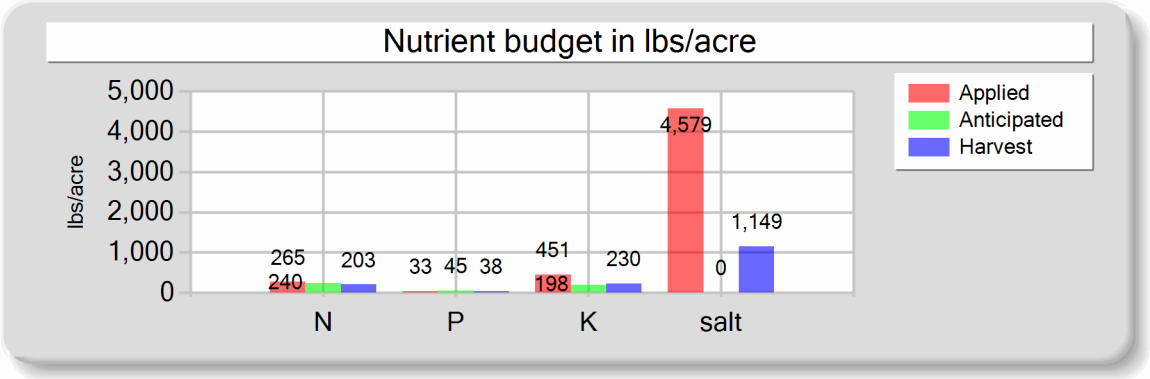
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,890,000.00	*gallons*
69.60	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/20/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.03	33.36	451.38	3,768.77
Fresh water	0.00	0.00	0.00	809.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.03	33.36	451.38	4,578.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	203.17	37.92	230.35	1,148.90
Nutrient balance	61.86	-4.56	221.03	3,429.78
Applied to removed ratio	1.30	0.88	1.96	3.99

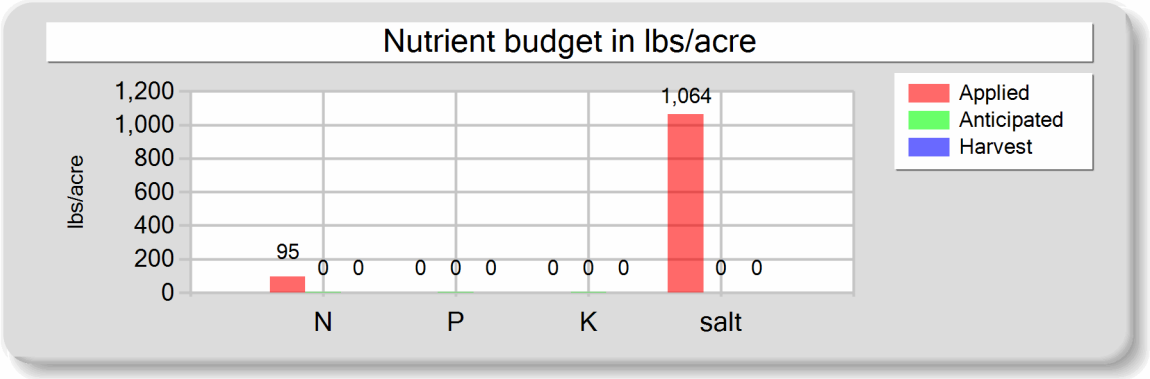
Fresh water applied	
33,968,700.00	*gallons*
1,250.95	*acre-inches*
29.78	*inches/acre*

Process wastewater applied	
2,688,000.00	*gallons*
98.99	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/20/2020: Pistachios

Field name: Field 3 Crop: Pistachios Plant date: 10/20/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25.50	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	55.80	0.00	0.00	1,063.70
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	95.30	0.00	0.00	1,063.70
Anticipated crop nutrient removal	0.10	0.01	0.01	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	95.30	0.00	0.00	1,063.70
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
21,571,000.00	*gallons*
794.39	*acre-inches*
36.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 19.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	5,700.00	17,700.00	16,000.00	5,500.00	6,000.00	3,100.00	1,350.00		74.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solds

Sample and source description: Corral Solds
Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 21.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,200.00	8,030.00	34,000.00							40.20
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1
Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	588.00	459.00	0.00	0.00	48.50	271.00	87.90	36.90	133.00	3,280.00	0.00	276.00	851.00	7,890.00	4,960
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.10	1.00	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	510.00	417.00	0.00	0.00	80.20	653.00								6,870.00	4,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	426.00	0.00	0.00	53.80	939.00								8,580.00	8,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	381.00	260.00	0.00	0.00	93.90	486.00								4,750.00	3,180
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW #1

IW #1

IW #1

Sample description: IW #1

Sample date: 04/22/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.82	0.00	4.02								308.00	130
DL	0.50	0.50	0.50								1.00	10

IW #2

IW #2

Sample description: IW #2

Sample date: 04/22/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								275.00	120
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 2 - 10/31/2022: Oats, silage-soft dough

Field 2 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,790.00	1,260.00	7,930.00		9.74
DL	100.00	100.00	100.00		0.01

Field 2 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,840.00	530.00	3,220.00		7.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

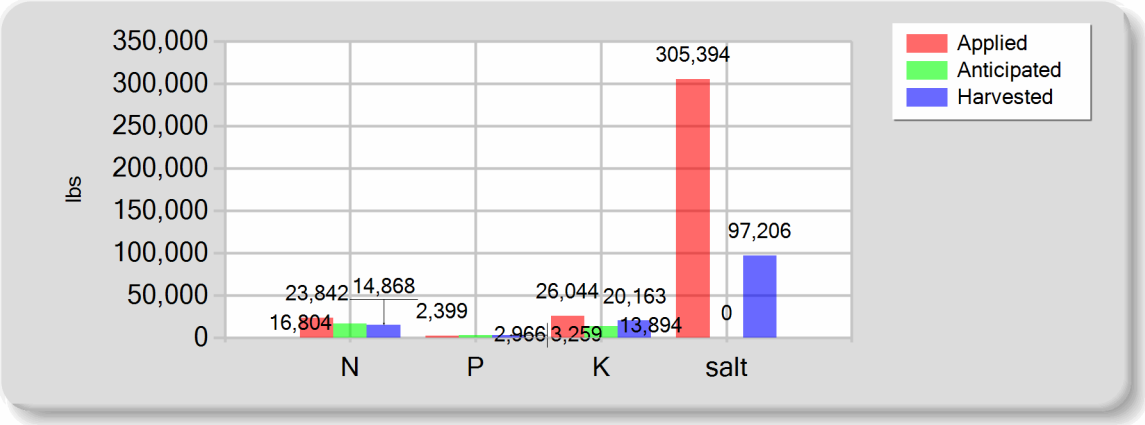
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

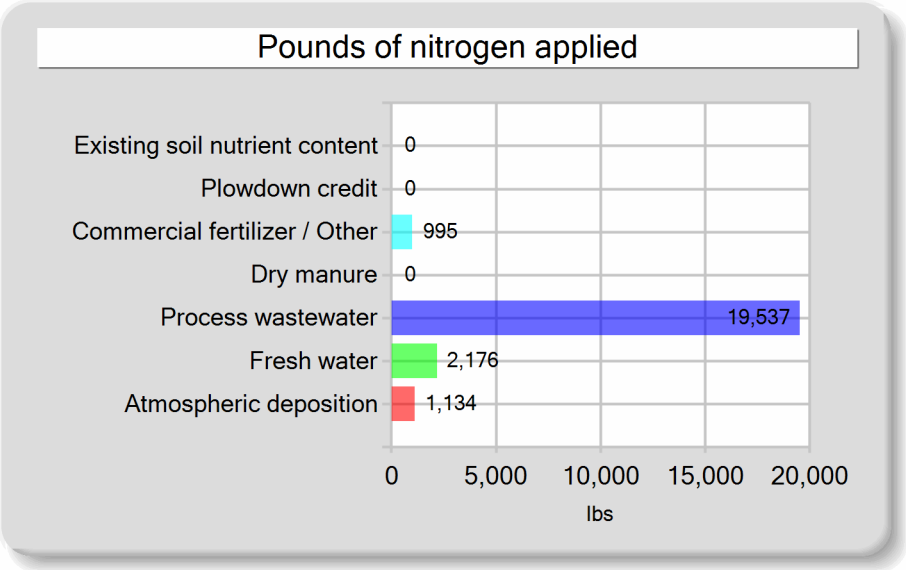
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

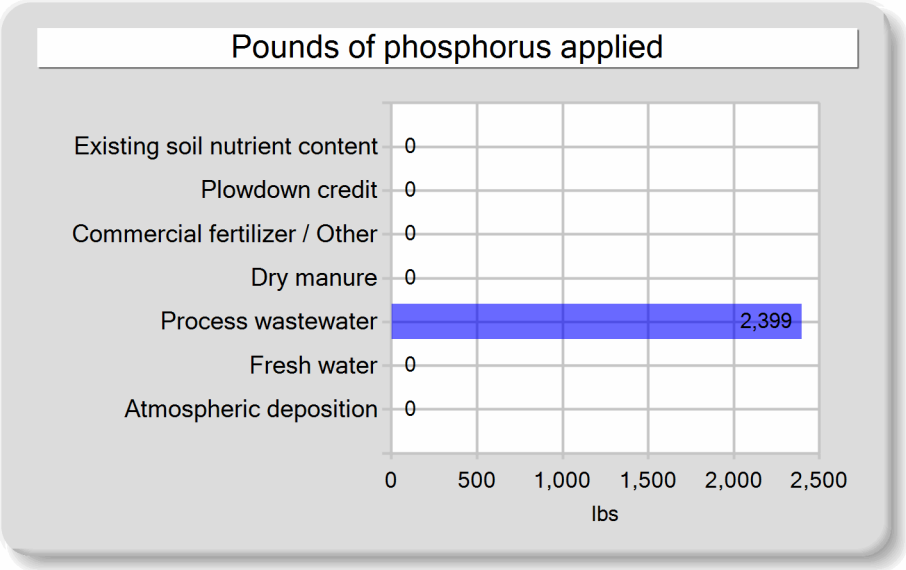
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	994.50	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	19,537.01	2,399.38	26,043.86	229,893.40
Fresh water	2,176.32	0.00	0.00	75,500.38
Atmospheric deposition	1,134.00	0.00	0.00	0.00
Total nutrients applied	23,841.83	2,399.38	26,043.86	305,393.78
Anticipated crop nutrient removal	16,803.90	2,965.59	13,893.99	0.00
Actual crop nutrient removal	14,868.32	3,258.91	20,163.03	97,205.95
Nutrient balance	8,973.52	-859.53	5,880.83	208,187.83
Applied to removed ratio	1.60	0.74	1.29	3.14

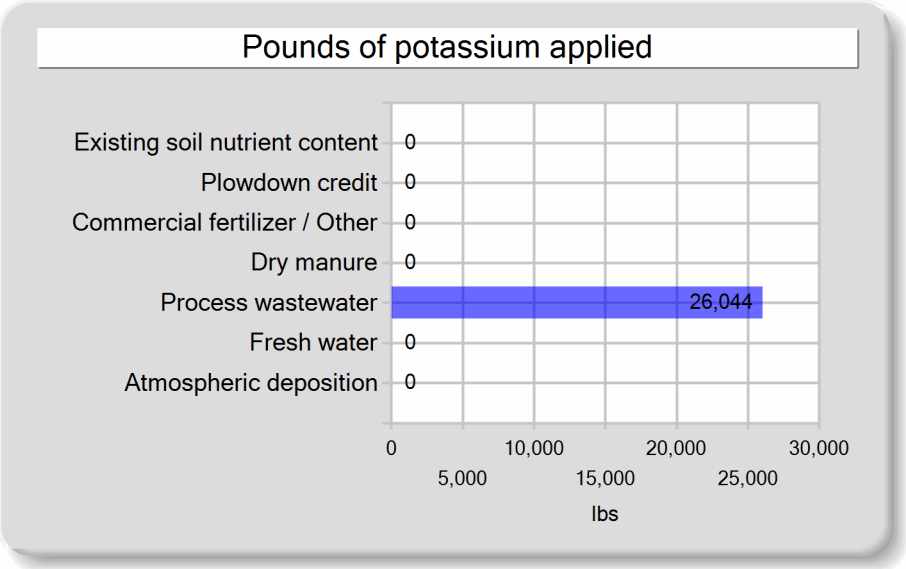
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

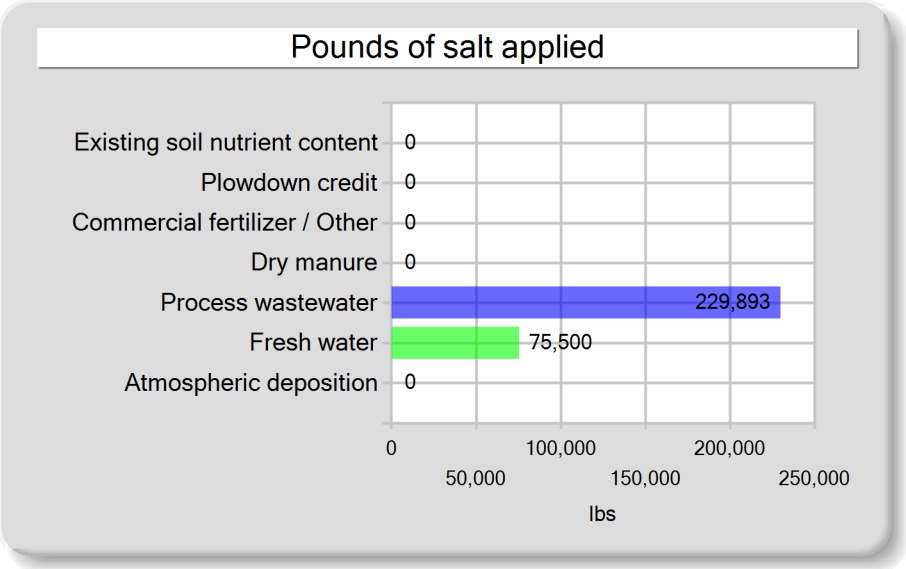


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 2 Oats (1.42) had a slightly high nitrogen field ratio as crop failed to meet anticipated nitrogen removal of 160 pounds/acre. The subject field removed 150.8 pounds of nitrogen/acre. Yield for the field was adequate however low nitrogen value from the plant tissue lab analysis resulted in 9.58 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton which ultimately resulted in the high nitrogen field ratio. Nitrogen application to the field was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Jarred Mello (May 6, 2024 12:10 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jarred Mello	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 6, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Jarred Mello

Name of Dairy Facility: Mello Ranch - Carter

Facility Address: 27181 Carter Road, Escalon, CA 95320
Number and Street City Zip Code

Contact Person Name and Phone Number: Jarred Mello 209-573-3748
Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person: ALP

Address of Hauling Company /Person: PO Box 218 Newman, CA 95360
Number and Street City Zip Code

Contact Person: Jeff Lima 209-862-3393
Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

J & R Sanguinetti Farms 18626 Copperopolis Road, Linden, CA 95236 209-462-6973
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:
18626 Copperopolis Road Linden, CA 95236 183-210-24
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 11/03/2023

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 564 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 21.7% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____
47 loads at 12 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Jarred Mello (May 6, 2024 12:10 PDT) Date: May 6, 2024

Hauler's Signature: _____ Date: _____

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Mello Ranch
P.O. Box 22
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Carter
RECEIVED DATE: 4/22/24 3:00 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 4/23/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Milk Barn	T4113023A	Electrical Conductivity (EC)	EPA 120.1 and	0.254	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.29	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	162	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	104 (T4113019A)	98.8 (T4113019A)	4.73
Electrical Conductivity (EC)	96.3			0.841
Nitrate Nitrogen (NO3-N)	99.1	104 (T4108024A)	105 (T4108024A)	0.821

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Mello Ranch
P.O. Box 22
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Carter
RECEIVED DATE: 4/22/24 3:00 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 4/23/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 1	T4113023B	Electrical Conductivity (EC)	EPA 120.1 and	0.308	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.02	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	130	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	6.82	mg/L
		Soluble Salts (SALT-SOL)	Calculation	197	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	104 (T4113019A)	98.8 (T4113019A)			4.73
Electrical Conductivity (EC)	96.3			0.955 (T4113019A)	0.947 (Dup)	0.841
Nitrate Nitrogen (NO3-N)	99.1	104 (T4108024A)	105 (T4108024A)			0.821
Total Dissolved Solids (TDS)	94.0			1200 (T4120026A)	1220 (Dup)	1.66
Total Nitrogen (TN)	101			2.80 (T4113023B)	2.80 (null)	0.000

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Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Mello Ranch
P.O. Box 22
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Carter
RECEIVED DATE: 4/22/24 3:00 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 4/23/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 2	T4113023C	Electrical Conductivity (EC)	EPA 120.1 and	0.275	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	120	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	176	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	104 (T4113019A)	98.8 (T4113019A)			4.73
Electrical Conductivity (EC)	96.3			0.955 (T4113019A)	0.947 (Dup)	0.841
Nitrate Nitrogen (NO3-N)	99.1	104 (T4108024A)	105 (T4108024A)			0.821
Total Dissolved Solids (TDS)	94.0			1200 (T4120026A)	1220 (Dup)	1.66
Total Nitrogen (TN)	101			2.80 (T4113023B)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Rush

Client: Mello / Cartes
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R Ag Services
Sampled By: Jeanette
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	4/22/24		DW Milk Barn	Domestic	23.1°C		✓				
2	↓		IW1	irrigation	23.9°C						H7
3	↓		IW2	↓	22.8°C						H7
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st	Jeanette Sanzer	F&R				T411302 3
2nd		DENELE ANALYTICAL, INC.	4/22/24 3:00			

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

2023 Annual Report - Mello Ranch Carter Road

Final Audit Report 2024-05-06

Created:	2024-05-06
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAABKjJ1c8SI-ADvmMoWa58cnbxdbSZHa5r

"2023 Annual Report - Mello Ranch Carter Road" History

Document created by Vince Furtado (vfurtado@fragservices.com)
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Signer jarredmello@hotmail.com entered name at signing as Jarred Mello
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